Securing the skill of using a pair of compasses

Task A: Drawing diagrams containing multiple circles

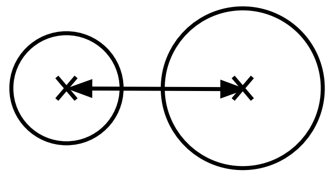
1) For each question, draw two circles: one with a radius of 5 cm and the other with a radius of 7 cm. Mark the positions of their centre points each time. Draw each pair of circles according to the following conditions:

Use a blank piece of A4 paper to make your drawings. They will not fit on this sheet.

To check, measure the distance between the two centre points in each question.

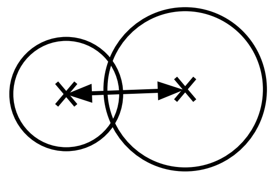
a) The circles do not touch. Neither circle has its centre point inside the other circle.

12 cm < distance



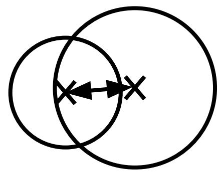
b) The circles intersect at two points. Neither circle has its centre point inside the other circle.

7 cm < distance < 12 cm



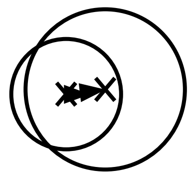
c) The circles intersect at two points. One circle has its centre point inside the other circle.

5 cm < distance < 7 cm



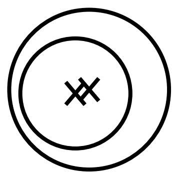
d) The circles intersect at two points. Both circles have their centre point inside the other circle.

2 cm < distance < 5 cm



e) The circles do not touch. Both circles have their centre point inside the other circle.

distance < 2 cm



Task B: Drawing circles that intersect exactly once

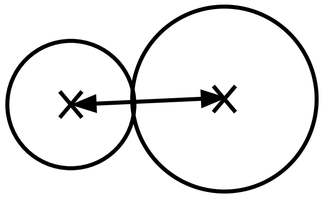
You will need blank A4 paper for Task B. Your drawings will not fit on this sheet.

1) For each question, draw two circles: one with a radius of 5 cm and the other with a radius of 7 cm. Mark the positions of their centre points each time. Draw each pair of circles according to the following conditions:

To check, measure the distance between the two centre points in each question.

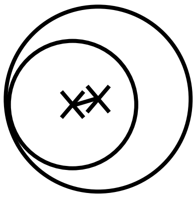
a) The circles touch exactly once. Neither circle has its centre point inside the other.

distance = 12 cm



b) The circles touch exactly once. One circle has its centre point inside the other.

distance = 2 cm



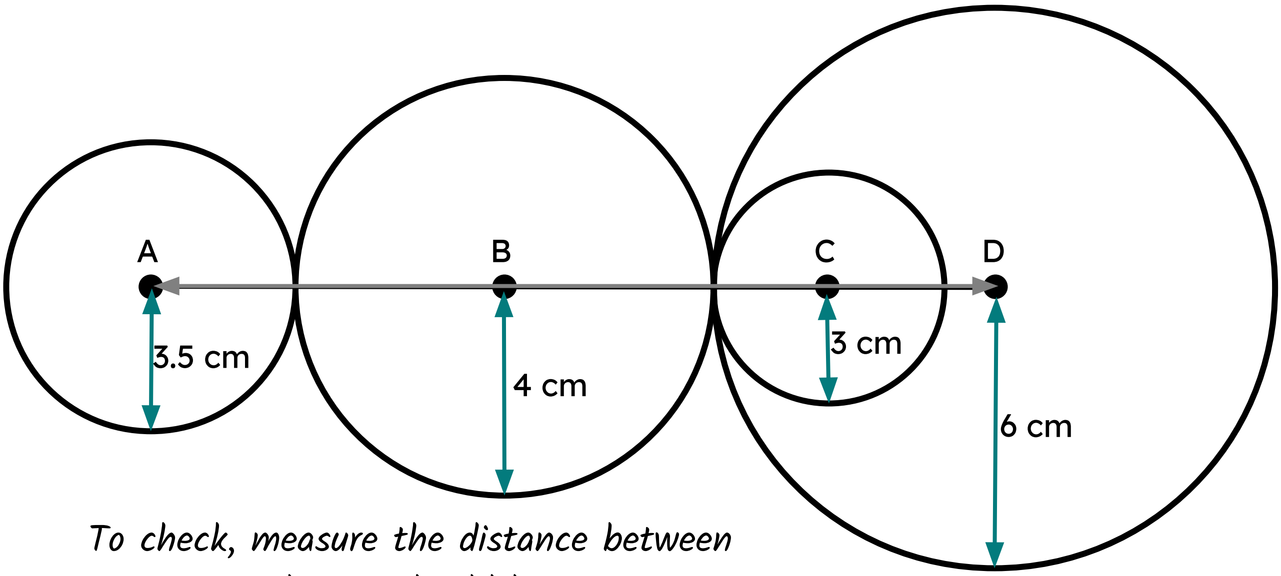
2) Draw two points 12 cm apart. Draw a circle around each point so that they touch each other exactly once. What is the radius of each circle?

Any pair of positive numbers that sum to 12 (e.g. 6 and 6).

or

Any pair of positive numbers with a difference of 12 (e.g. 14 and 2).

3) Points A, B, C and D are on the same line. Draw an accurate version of the diagram.



To check, measure the distance between points A and D. It should be 17.5 cm.